

# Antenna Switch Project

## Summary

The system is intended to be a remotely controlled antenna switching and tuning system to be mounted outdoors in a weatherproof junction box. The system is intended to operate in the 1-30 MHz range at transmitting power levels of 100W. The master controller is located at the radio operator's position and communicates to the remote parts of the system via an RS485 communications link. The remote system contains 3 active elements: a custom antenna switching relay array, an ATU-100 antenna tuner, and a remote controller based on a Pico2W microcontroller. The intended functionality is to enable the operator to select one of 4 antennas and run the ATU-100 as needed using its existing functionality.

## Master Controller

Based on a RaspberryPi Pico2W microcontroller (programmed in MicroPython) augmented with an I/O expansion board (Adafruit MCP23017 I2C GPIO Expander) and an RS485 module ( M5Stack RS485- ISO Unit – Isolated RS485 Transceiver P/N: U094). The user interface is a 4-line 128x64 pixel OLED display (I2C interface) and a set of 10 push buttons and 5 LEDs. Display will mimic the display on the ATU-100 using data captured by the remote controller and communicated to the master via the RS485 communications link. Commands can be sent to the remote controller via RS485 to select the antenna to be used and to enable ATU-100 functions as needed.

## Remote Controller

The remote controller is also based on a RaspberryPi Pico2W microcontroller (programmed in MicroPython) augmented with the same I/O expansion board and RS485 module as the Master Controller. The user interface is the same 4-line 128x64 pixel OLED display (I2C interface) and a set of 10 push buttons and 5 LEDs. The remote controller is housed in a metal enclosure and isolated from RF circuits and CMC interference.

## Antenna Switch

The antenna Switch is a bank of four relays, each controlled by a MOSFET switch driven by an optically isolated input from the Remote Controller's I/O. Thus, there are four relay

channels and four I/O control lines required. All relays are off (i.e. not energized) unless one is selected for use. Only one relay should be activated at any one time.

## ATU-100 Antenna Tuner

The ATU-100 Antenna Tuner uses a PIC16F1938 as its control processor. Its software is written in C and is available on GitHub at <https://github.com/Dfinitiski/N7DDC-ATU-100-mini-and-extended-boards>. The current firmware version is 3.2 and is specifically for the ATU\_100\_EXT\_board, which is used in this project. The current user manual is located here: [https://github.com/Dfinitiski/N7DDC-ATU-100-mini-and-extended-boards/tree/master/ATU\\_100\\_EXT\\_board/ATU\\_100\\_EXT\\_board\\_User\\_Manual](https://github.com/Dfinitiski/N7DDC-ATU-100-mini-and-extended-boards/tree/master/ATU_100_EXT_board/ATU_100_EXT_board_User_Manual).

## Functional goals:

- The Master and Remote controllers can communicate data and commands via a half-duplex RS485 link using a frame based protocol containing a data packet and using a check sum for validation.
- All I/O and data states are maintained in a state table. The current state table reflects the I/O and data states after a new command is received. A previous state table is maintained as a record of the I/O and data states prior to any change. This will allow any error resulting from a state change event to revert to the previous state as recorded in the previous state table.
- The antenna relays are maintained in an off state (i.e. not actuated) unless activated for use. Only one relay at a time can be activated. The active relay must be deactivated with a small delay to allow deactivation to complete. Then the newly selected relay is driven to an on state (i.e. actuated). [**Note:** There is no sensor feedback to show that a given relay has been successfully activated or deactivated.]
- The ATU-100 uses an SWR sensor to determine the forward and reflected power on the transmission line. It uses relays to switch different inductor (L) and capacitor (C) values into the RF circuit path to minimize the forward to reflected power ratio as close to 1:1 as possible. The settings that yield the lowest SWR reading allow the highest efficiency of power transfer from the RF source (i.e. transmitter) to the RF load (i.e. antenna). An internal Tuning Algorithm in the ATU-100 firmware is in control of this process. Data is also captured and reconfigured for rendering on the ATU-100 OLED display. Data displayed includes the following:
  - Forward Power in Watts as a decimal number and one decimal place with the terminal letter "W" indicating the units in Watts,

- SWR as a decimal number to two decimal places with no units,
  - The selected L value (inductance) in decimal and two decimal places with the terminal letters “uH” to represent units in microhenrys,
  - The C value in decimal with no decimal places and with the terminal letters “pF” to represent units in picofarads,
  - In some circumstances (which are not clear from the user instructions) after a Tune sequence occurs, the display will report Forward Power, SWR, Power at the Antenna, and an efficiency rating in percent efficiency. Please see the ATU-100 code for further information.
- The Tune/Reset button initiates the ATU-100 to perform a tuning cycle per the tuning algorithm (it is connected to RB0 on the PIC16F1938),
  - The ATU-100 has an “automatic mode” or “Manual Mode;” The mode is selected by the “A/M” button, which toggles between the two states, (it is connected to RB1 on the PIC1938)
  - There is a Bypass button (it is connected to RB2 on the PIC16F1938) that causes the RF input to be routed directly to the RF output with all L and C values deselected. When in bypass mode the display shows an underscore character at the end of the first line. When in Bypass Mode, all capacitors and inductors are removed from the RF circuit path, which means that no tuning is being done.
  - When in automatic mode, a period symbol will appear at the end of the first line of the ATU-100 display.
  - When power is initially turned ‘ON’ the ATU-100 defaults to the manual mode,
  - In the automatic mode, the tuning algorithm is triggered when the SWR is greater than 1.3:1.
  - When the A/M button and the Bypass button are held down and power is applied to the ATU-100, it enters a “Test mode.” In Test mode, the A/M and Bypass buttons can be used to manually step through the available capacitance and inductance values. The Tune button changes whether the capacitors or inductors are being changed. Further details about this functionality and any display changes should be taken from the ATU-100 code itself.
  - Only two I/O lines are unused on the PIC16F1938: RA7 and RA6. These two lines are to be used for the communication link with the remote controller. A bit banging protocol should be used where the ATU-100 acts as the master and the Pico2W acts as the slave or listener. When a change in any operational state occurs, it should be reported to the slave. However, the slave needs to be able to alert the master when a command is ready to be sent that has originated from the Master Controller. The

protocols here are not yet defined beyond that, but should include inquiry, ready to send, ready to receive, data packet transfer, authentication, acknowledgement, etc.

- Bit Banging commands

Note: This is an initial draft and is subject to change as needed

| From ATU-100 to Remote Controller | To Remote Controller from ATU-100 | To Master Controller from Remote Controller | To Remote Controller from Master Controller |
|-----------------------------------|-----------------------------------|---------------------------------------------|---------------------------------------------|
| ACK (acknowledge)                 | ACK (acknowledge)                 | ACK (acknowledge)                           | ACK (acknowledge)                           |
| HHH (Hello, Are you there?)       | HHH (Hello, Are you there?)       | HHH (Hello, Are you there?)                 | HHH (Hello, Are you there?)                 |
| RR (Ready to receive)             | RS (Ready to Send)                | RPT (Repeat last message)                   | RPT (Repeat last message)                   |
| STA (Status Report Follows)       | STA (Status Data Packet follows)  |                                             |                                             |
| RS (Ready to Send)                | RR (Ready to receive)             |                                             |                                             |
|                                   |                                   |                                             |                                             |
|                                   |                                   | ACK AT1(send after execution of command)    | AT1 (Activate Ant Relay 1)                  |
|                                   |                                   | ACK AT2 (send after execution of command)   | AT22 (Activate Ant Relay 2)                 |
|                                   |                                   | ACK AT3 (send after execution of command)   | AT3 (Activate Ant Relay 3)                  |
|                                   |                                   | ACK AT4 (send after execution of command)   | AT4 (Activate Ant Relay 4)                  |
|                                   |                                   | ACK AT0 (send after execution of command)   | AT0 (Deactivate all Ant relays)             |
|                                   |                                   | RR (Ready to receive)                       | RS (Ready to Send)                          |
|                                   |                                   | RS (Ready to Send)                          | RR (Ready to receive)                       |
|                                   |                                   |                                             |                                             |
|                                   |                                   |                                             |                                             |
|                                   |                                   |                                             |                                             |
|                                   |                                   |                                             |                                             |

| From Master Controller to Remote Controller | From Remote Controller to Master Controller | From Remote Controller to Antenna Switch | From Remote Controller to ATU-100 | From ATU-100 to Remote Controller |
|---------------------------------------------|---------------------------------------------|------------------------------------------|-----------------------------------|-----------------------------------|
| AT0 (Deactivate all Ant relays)             | ACK AT0                                     | Deactivate all AT relay I/O lines        |                                   |                                   |
| AT1 (Activate Ant Relay 1)                  | ACK AT1                                     | Activate AT1 relay I/O line              |                                   |                                   |
| AT2 (Activate Ant Relay 2)                  | ACK AT2                                     | Activate AT2 relay I/O line              |                                   |                                   |
| AT3 (Activate Ant Relay 3)                  | ACK AT3                                     | Activate AT3 relay I/O line              |                                   |                                   |
| AT4 (Activate Ant Relay 4)                  | ACK AT4                                     | Activate AT4 relay I/O line              |                                   |                                   |
| TUN (Tune)                                  | ACK TUN                                     |                                          | Send Tune Command                 |                                   |
| BYP0 (Clear Bypass)                         | ACK BYP0                                    |                                          | Clear Bypass                      |                                   |
| BYP1 (Set Bypass)                           | ACK BYP1                                    |                                          | Set Bypass                        |                                   |
| AM0 (Set Auto Mode)                         | ACK AM0                                     |                                          | Set Auto Mode                     |                                   |
| AM1 (Set Manual Mode)                       | ACK AM1                                     |                                          | Set Manual Mode                   |                                   |
| TST0 (Clear Test Mode)                      | ACK TST0                                    |                                          | Clear Test Mode                   |                                   |
| TST1 (Set Test Mode)                        | ACK TST1                                    |                                          | Set Test Mode                     |                                   |
| TUP (Manual Value Up 1 step)                | ACK TUP                                     |                                          | Step Up (A/M button Press)        |                                   |
| TDN (Manual Value Down 1 Step)              | ACK TDN                                     |                                          | Step Down (BTP button press)      |                                   |
| TSC (Test Select Capacitors)                | ACK TSC                                     |                                          | Select C (Tune button press)      |                                   |

| From Master Controller to Remote Controller                                                         | From Remote Controller to Master Controller | From Remote Controller to Antenna Switch | From Remote Controller to ATU-100    | From ATU-100 to Remote Controller  |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------|--------------------------------------|------------------------------------|
| TSL (Test Select Inductors)                                                                         | ACK TSL                                     |                                          | Select L (Tune button press)         |                                    |
|                                                                                                     | ERR (Error ##)                              |                                          |                                      |                                    |
|                                                                                                     |                                             |                                          | SND (Send Data Packet)               | Status Data Packet Ready           |
|                                                                                                     |                                             |                                          | RCVD (Data Packet Received)          | Send Data Packet                   |
|                                                                                                     |                                             |                                          | RPT (Send Data Packet Again)         | ACK                                |
|                                                                                                     |                                             |                                          |                                      | Send Data Packet                   |
| F86 (Shut Down all Hardware)                                                                        |                                             |                                          |                                      |                                    |
| RST (Reset)                                                                                         | ACK RST                                     |                                          | RST (Reset to normal power up cycle) |                                    |
| RST RDY (Power on reset and ready notice)                                                           | RST RDY (Power on reset and ready notice)   |                                          |                                      | RST RDY (Power on reset and Ready) |
| STA (If following a RST RDY command, this is a request for an ops status on the remote controller.) |                                             |                                          |                                      |                                    |
|                                                                                                     |                                             |                                          |                                      |                                    |
|                                                                                                     |                                             |                                          |                                      |                                    |

- Data received by Remote Controller from the ATU-100 is packaged and sent to the Master Controller where the data is unpacked and a status table is filled with the data. Data from the status table is then displayed on the Master Controller's display, mimicking the ATU-100's display.

## RS485 Data Frame Structure

Note: The following is only an initial design and may be changed as necessary.

Open TBD.

## Embedded Controller Hardware

The Master Controller and the Remote Controller will have the same basic hardware configuration.

- The microcontroller will be the RaspberryPi Pico 2W board.
- The RS485 module will be the M5Stack RS485-ISO Unit – Isolated Transceiver Model U094. It will connect to the Pico 2W UART port.
- The I/O expansion module will be the Adafruit MCP23017 I2C GPIO Expander that connects to the Pico 2W via an I2C port. The default I2C address for this board is 0x20. Based on the MCP23017 chip, the I/O Expansion board provides for 16 bidirectional I/O logic pins in 2 ports (A0-A7 and B0-B7) along with an interrupt pin for each port (INTA and INTB). There is also a reset pin. It also has an I2C interface. There are also two STEMMA ports.
- The OLED display (128×64 pixels) also connects to the Pico 2W using an I2C (4-wire connection: GND, VCC, SCL, SDA) port. I2C Address: 0x3C (7-bit) or 0x78 (8-bit write address). It uses the SSD1306 as its driver IC and communicates via I2C interface. It is 3.3–5V compatible. Functional capabilities should include:
  - Rendering full lines of text on specific lines 1-4, with additional lines buffered for scrolling by command.
  - Rendering individual characters by specifying line number and position number.
  - Facilitating the depiction of a scrollable menu system. The current item should be highlighted, which also represents the cursor position. An “Exit Menu” option will terminate the menu and return to the normal display. Selecting a submenu option will also exit from the menu system as well as initiating a handling routine for the sub menu selection. Note that 5 buttons will be used by the menu system:

- 1 button scrolls up
- 1 button scrolls down
- 1 button scrolls left
- 1 button scrolls right
- 1 button acts as a select button

Both controllers will include a user interface consisting of the OLED display and several buttons for input.

Where the two controllers differ is that the Master Controller will only have to deal with the RS485 module, the OLED display and up to 12 button inputs using the I/O expansion board. The Remote Controller will be dealing with a basic user interface similar to the Master (probably with fewer buttons) as well as the RS485 and I/O Expansion modules. But it will differ in that it will be supporting a 2-wire interface to the ATU-100 and four I/O output lines that are to be used for Antenna Relay Control.

**[Note:** If it is not possible to use the bit banging interface to trigger button input functions in the ATU-100, then additional I/O lines from the Remote Controller will be used to connect in parallel with the physical button wiring in the ATU-100.]

## Software

The software design is non-existent at this point. However, I would like to have a system that uses a continuous command loop to execute different “tasks” included in the loop. One of those tasks would have to be an Interrupt handler that can respond to hardware driven interrupts, possibly with multiple instances included in the command loop so that interrupts would be handled in a timely manner. “Tasks” would include higher level command sequences to implement control of data handling tasks.

The state of I/O inputs and outputs as well as discrete data needed for command and control functions would be maintained in a state table that would be updated and executed whenever a new command or control datum occurred. Two state tables would be used: a current state table would be populated with the current command and control data. A second table would contain the same data points but reflect the previous state. The previous state table would be used for error correction if there was a problem with the data in the current state table or an error indicating that the previous state table values should be used as a safety fall back.

The ideal outcome here is if all functionality currently implemented in the ATU-100 code was accessible to the Remote Controller. In this way, the ATU-100 would become a fully functional remote antenna tuner. The Master Controller would receive data downloads normally fed into the ATU-100 display and would be able to send operational commands to the ATU-100 emulating its button control inputs.

## Coding Languages

Code for the Pico 2W controllers should be implemented in MicroPython. The ATU-100 firmware code is currently written in C for the PIC16F1938.

## Error Handling

If errors occur from any command and control functions, the device experiencing the error should report the error up through the command chain. Thus, If the ATU-100 experiences an error, it should be reported to the Remote Controller (if possible), which would in turn report the error to the Master Controller. If possible, the source and nature of the error should be reported. System level errors are the most critical. Local errors within one of the controllers should be alerted to the display. Typical errors would include:

- Failed to Execute Command
- Data Not Available
- Resource offline
- Communication Lost with unit xxxx
- Data Corrupted from xxxxx
- Etc.